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Dual-Channel Graph Ranking Contrastive Learning for Dissolved Gas Analysis

Yingyue Zhang, Huifang Ma, *Member, IEEE*, Di Zhang, Ke Shu, and Xiaolong Li

***Abstract*—Dissolved Gas Analysis (DGA) is a widely used diagnostic technique specifically designed to identify potential defects or irregularities in power transformers. Despite recent advancements, the fine-grained correlation modeling between samples is still insufficient. This limitation results in less contextualized feature representations, ultimately impacting the accuracy of diagnosis. To this end, we propose an effective graph deep learning model called DGRCL (Dual-channel Graph Ranking Contrastive Learning) for DGA. In our model, we construct two types of graphs: similar K*-Nearest Neighbors (*KNN) (G_S) and dissimilar *KNN (G_{DS}), where nodes represent individual records of dissolved gases and the edges, namely similar-homophilic and dissimilar-heterophilic, capture the relationships between nodes. To better generate node representations, we design a dual-channel graph encoding component, with each channel playing a crucial role in capturing the corresponding edge type view. Furthermore, in order to better distinguish the feature differences among various fault categories, we employ ranking contrastive loss to refine the node representations. Experimental results demonstrate that our model achieves high accuracy and robustness in predicting fault types.**

***Index Terms*—Dissolved Gas Analysis (DGA), Graph Encoding, Power Transformer, Ranking Contrastive Loss.**

I. Introduction

POWER Transformer Fault Diagnosis (PTFD) [1] refers to the process of detecting, analyzing, and assessing the operational status to determine the presence of faults or abnormalities. Furthermore, it aims to identify the type and location of the fault. Effective and reliable transformer fault diagnosis enables the timely detection and localization of faults, aiding maintenance personnel in choosing the correct repair measures to ensure the normal operation of transformers. However, real-time monitoring of transformers often requires a significant amount of human and material resources. Therefore, employing computational methods for efficient and accurate PTFD rather than relying solely on on-site inspections can enhance the efficiency of PTFD.

Due to its high practical value, numerous effective methods have been developed in recent years for PTFD [2], [3], [4]. Among these, Dissolved Gas Analysis (DGA) is a commonly used diagnostic technique in the power industry, which effectively improves fault detection reliability and operational safety of transformers. This approach involves regular collection of transformer oil samples and analyzing the concentrations of dissolved gases, such as hydrogen (H2), methane (CH4), ethane (C2H6), ethylene (C2H4), and acetylene (C2H2). When internal faults or abnormal conditions occur within the transformer, such as arc discharge, partial discharge (PD), overheating, or other irregularities occur within the transformer, specific gases are dissolved in the insulating oil. Consequently, DGA has become a promising method for transformer fault diagnosis. With the widespread use of machine learning, recent endeavors have amalgamated DGA with various deep learning techniques, including Support Vector Machines (SVM) [5], K-Nearest Neighbors (KNN) [6], Decision Trees (Tree) [7], and Back Propagation Neural Networks (BPNN) [8], aiming to augment model detection accuracy. Nevertheless, the incapacity of these methods to capture the intricate relationships among existing samples results in inadequacies in the modeled features of DGA data, thereby constraining the performance of transformer fault diagnosis.

Graph Neural Networks (GNNs) have garnered significant attention and have been applied across various domains. For instance, in knowledge tracing [9], a student's historical interactions are modeled as a graph, where GNNs predict the likelihood of correct answers to assess their knowledge state. In recommendation systems [10], both users and items are represented as nodes, with GNNs facilitating effective representation learning. Likewise, in disease similarity prediction [11], diseases are modeled as nodes, allowing GNNs to uncover complex relationships among them. Inspired by these successful applications, we propose a novel approach that represents individual samples as nodes within a graph, leveraging GNNs' powerful representation learning capabilities to capture deep underlying structures. However, several challenges still need to be addressed. *1) How to construct appropriate graphs in a reasonable manner?* Many existing methods merely rely on the original features of DGA data for transformer fault detection, failing to capture the complex relationships between samples. Thus, determining how to construct a well-designed and effective graph, along with selecting suitable hyperparameters, is of utmost importance

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for accurate DGA. *2) How to deal with the issue of blurred boundaries between fault categories?* Different fault types can lead to the generation of various gases, and in practice, gas concentrations may overlap to some extent, making it difficult to define clear boundaries between fault types. Expanding and refining the boundaries between different fault categories within the DGA data are crucial for achieving effective transformer fault detection.

To address the aforementioned challenges, we propose a graph contrastive learning model called **D**ual-channel **G**raph **R**anking **C**ontrastive **L**earning (DGRCL). DGRCL consists of three main modules: graph representation learning module, ranking contrastive learning module and fault diagnosis module. The first module exploits the unique characteristics of DGA data. It constructs two types of KNN graphs: one with similar-homophilic edges and the other with dissimilar-heterophilic edges. A dual-channel graph encoding mechanism is then introduced to learn node representations on these different graphs, capturing the complex relationships within the DGA data. The second module is specifically designed to address the issue of blurred boundaries between fault categories in DGA data. It optimizes the model through ranking contrastive learning [12] paradigm. This module focuses on finely ranking the similarities between fault categories, enabling the model to differentiate the fault categories accurately based on their similarities. The last module utilizes the embedding of optimized nodes (samples) for fault diagnosis.

The main contributions of this article are as follows:

Findings: We have conducted data analysis and made the following discoveries: 1) Constructing two types of graphs for DGA proves to be highly beneficial as it significantly enhances fault detection. 2) When considering fault categories, it is observed that the similarity within the same category, such as discharge-type faults, is notably higher compared to the similarity between fault categories caused by different factors.

Methodology: Inspired by the insights gained from the data analysis, we propose a multirelational graph contrastive learning network, DGRCL. This model utilizes a dual-channel graph encoding component to learn node representations from different graphs and introduces ranking contrastive loss to differentiate feature differences between various fault categories. To the best of the authors knowledge, this is the first work employing multirelational graph contrastive learning to address transformer fault detection.

Experiments: Experimental results on two widely used DGA datasets demonstrate that DGRCL surpasses state-of-the-art DGA methods. Furthermore, various analyses underscore the substantial contributions of each component of DGRCL to the final performance, affirming the interpretability and generalization of DGRCL.

The rest of this article is organized as follows. Section II introduces the commonly used symbols in DGRCL, and provides a thorough analysis of the data from the perspectives of graph construction and fault boundaries. Section III presents a detailed explanation of the specific implementation of DGRCL. Section IV validates the effectiveness of DGRCL on two public datasets and examines its generalization ability and robustness. Finally, Section V concludes this article and discusses possible future research directions.

II. Preliminaries

A. Notations

Let $\mathbf{X} \in R^{n\times d}$ denote the original feature matrix containing n samples of dissolved gas in oil, with each sample represented as a d dimensional vector $\mathbf{x}_i$. We then construct two distinct graphs based on $\mathbf{X}$: a similar KNN graph $G_S = (V, \mathbf{A}_S)$ and a dissimilar KNN graph $G_{\text{DS}} = (V, \mathbf{A}_{\text{DS}})$. Both graphs share the same node set while possessing different graph structures, as explained in detail in Section II-B. $\mathbf{A}_S$ and $\mathbf{A}_{\text{DS}}$ represents the weighted adjacency matrix of G_S and G_{DS}, $\tilde{\mathbf{A}}_S$ is the Laplacian normalization matrix of $\mathbf{A}_S$. $\mathbf{A}_{\text{DS}}$ and $\tilde{\mathbf{A}}_{\text{DS}}$ are calculated in the same manner as $\mathbf{A}_S$ and $\tilde{\mathbf{A}}_S$. ε_s and ε_{ds} denote the edge sets of graphs G_S and G_{DS}, respectively. $\mathbf{H}$ represents the node embedding matrix learned by DGRCL. The predicted probability vector $\hat{\mathbf{y}}_i$ represents the likelihood of sample i belonging to each fault class.

We formally state the transformer fault diagnosis task as follows.

- *Input:* The original DGA feature matrix of transformer samples, a subset of training samples with the ground truth labels $\mathbf{Y}$.
- *Output:* A prediction function that estimates the probability vector $\hat{\mathbf{y}}_i$ for each sample in the test set.

B. Graph Construction

Our work utilizes two reliable DGA datasets, namely, DGA-200[1] and DGA-2000.[2] The detailed statistics about the dataset are illustrated in Section IV, where comprehensive information is summarized. In this section, we analyze the effectiveness of graph construction using the aforementioned datasets.

Given a set of dissolved oil samples $\mathbf{X}$, we expect to construct a graph where each sample is represented by a node and two samples are linked by an edge if they are similar. Cosine similarity measures relationships between samples based on the angular alignment of their feature vectors, capturing nonlinear relationships without relying on linear assumptions. Previous studies [13], [14] show that connecting each sample to its top-KNN preserves graph connectivity and reveals hidden links between isolated samples. Therefore, we construct a G_S by connecting each sample with its top-K similar samples based on the cosine similarity as follows:

$$\mathbf{A}_{i,j} = \frac{\mathbf{x}_i^T \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \, \|\mathbf{x}_j\|} \tag{1}$$

where $\mathbf{A}_{i,j}$ denotes the weight of the edge connecting sample i and sample j, $\mathbf{x}_i^T \cdot \mathbf{x}_j$ denotes the inner product of vectors in the i-th and j-th rows of matrix $\mathbf{X}$, $\|\cdot\|$ represents the norm of the vector.

[1][Online]. Available: https://IEEE-dataport.org/documents/dissolved-gas-data-transformer-oil-fault-diagnosis-power-transformers-membership-degree

[2][Online]. Available: https://github.com/alan-456/transformer-fault-dataset

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TABLE I
PERFORMANCE COMPARISON OF GNNS BASED ON G_s AND G_{DS} WITH CLASSIC MACHINE LEARNING CLASSIFICATION METHODS

Methods	DGA-200		DGA-2000	
	AUC	F1	AUC	F1
BPNN	0.72±0.07	0.63±0.10	0.79±0.03	0.59±0.03
SVM	0.75±0.03	0.65±0.05	0.79±0.01	0.73±0.01
KNN	0.76±0.03	0.68±0.06	0.83±0.01	**0.79±0.02**
Tree	0.77±0.03	**0.70±0.07**	0.78±0.02	0.73±0.03
GNN(w/G_S)	**0.89±0.02**	0.66±0.05	**0.91±0.01**	0.73±0.02
GNN(w/G_{DS})	0.86±0.02	0.60±0.04	0.87±0.01	0.66±0.01

The best and second-best methods are highlighted in bold and underlined, respectively.

Previous studies have confirmed that indiscriminate aggregation of neighbor nodes by GNNs may lead to all node representations being close to each other, hindering model performance. This problem, known as over-smoothing, is prevalent in GNNs. To mitigate this issue, we introduce the dissimilar-KNN graph G_{DS} to provide a complementary perspective. Essentially, the target sample is connected to K samples with the lowest similarity to form heterophilic edges.

To investigate whether graph construction can alleviate the issue of incomplete feature learning, we conduct a series of preliminary experiments. We apply classical low-pass and high-pass GNNs [15] separately on graphs G_S and G_{DS}, and evaluate their performance in fault classification with the learned representations. As comparison methods, we select representative fault detection algorithms, including BPNN, SVM, KNN, and Tree. The experimental results are presented in Table I, which show the average and standard deviation of area under the curve (AUC) and F1 over 10 random splits of the dataset. GNN(w/G_S) and GNN(w/G_{DS}) methods indicate the results obtained with only G_S or G_{DS} for representation learning and classification. Detailed descriptions of hyperparameter settings are provided in Section IV.

From Table I, it can be concluded that the performance of GNN(w/G_S) and GNN(w/G_{DS}) is either optimal or suboptimal in most cases. This suggests that utilizing graph construction for DGA can indeed improve detection performance to some extent. However, the F1 score performance is suboptimal on both datasets, it indicates the necessity for further model design to enhance performance. Therefore, combining both graphs to learn shared attributes and differential information between nodes will mitigate the information loss issue introduced by single-channel graph networks.

C. Fault Category Similarity Analysis

Intuitively, the similarity between different fault types is expected to be low. To validate this assumption and formulate effective strategies for managing the relationships between fault types, we conduct a detailed analysis of the similarity levels across fault categories. By calculating the mean cosine similarity between all samples belonging to any two fault categories, we can measure the similarity between fault categories, as shown in Fig. 1. In the figure, red indicates higher correlation, while blue suggests lower correlation.

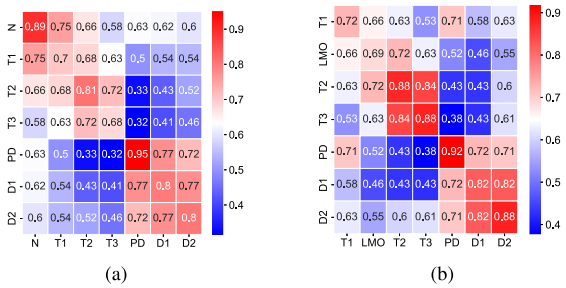


Fig. 1. Generated heatmap displays the similarity between fault categories. The numbers represent the average cosine similarity across all samples pairs of fault categories. (a) Fault category similarity heatmap on DGA-2000. (b) Category similarity heatmap on DGA-200.

Upon examining Fig. 1, several key observations can be made. First, within the DGA dataset, the highest degree of similarity is evident among faults that fall under the same category, as indicated by the darkest red shading along the diagonal of the figure. Second, it is apparent that the similarity among faults resulting from a shared underlying factor, such as temperature-induced faults T1, T2, and T3, is typically greater than the similarity between faults caused by different factors; for instance, comparing temperature-induced faults with discharge-induced faults, such as PD, D1, and D3. Lastly, in Fig. 1(a), the similarity scores for most pairs of faults exceed 32% , while in Fig. 1(b), these scores are above 38% . This suggests that the distinctions between different fault categories within the DGA dataset are not sharply defined. Consequently, failing to adjust the boundaries for categories with high similarity could potentially lead to an elevated risk of misdiagnosis during the diagnostic procedure

To sum up, the finding reveals that the high similarity among DGA fault categories leads to ambiguous classification boundaries. This indicates that different fault types can result in similar gas abnormal patterns. Consequently, accurately determining specific fault types becomes more challenging due to the fuzzy boundaries that make it difficult to distinguish between different faults.

III. METHODOLOGY

In light of the two findings discussed in the previous section, we introduce DGRCL, as depicted in Fig. 2. DGRCL consists primarily of three components: 1) Dual-channel representation learning module: This module incorporates a dual-channel encoding component. It applies separate low-pass and high-pass graph convolutions to the G_S and G_{DS} graphs. The resulting fused representation is considered as the final representation of the nodes. 2) Ranking contrastive learning module: This module leverages the hierarchical relationships among different fault categories. It optimizes the node representations by considering the ranking contrast between fault categories. 3) Fault diagnosis module: This module utilizes the optimized representations obtained from the ranking contrastive learning module to detect transformer fault categories.

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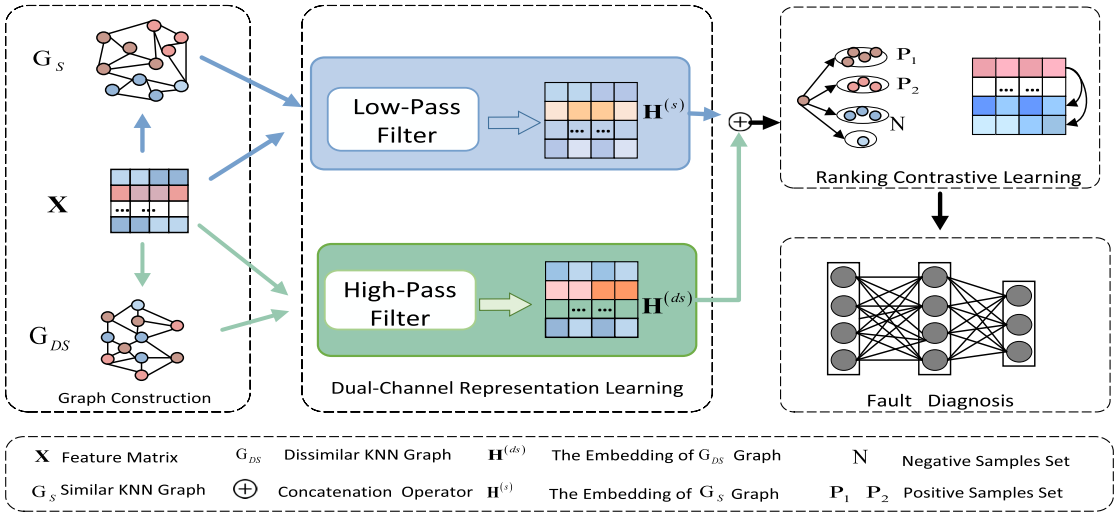


Fig. 2. Overall framework of DGRCL.

A. Dual-Channel Representation Learning

In order to capture comprehensive contextual information in DGA data, the model is endowed to learn both the shared features among similar samples and the distinctive features among dissimilar samples. Thus, DGRCL incorporates a dual-channel representation learning module, as shown in the first part of Fig. 2. By utilizing both low-pass and high-pass filters, DGRCL ensures that the model can effectively leverage both the shared and differing features in the DGA data, thus enhancing its ability to accurately detect and classify transformer faults.

1) Low-Pass Filtering: As previously discussed, modeling G_S proves beneficial in capturing the shared features among samples exhibiting the same transformer fault. In this module, we construct a low-pass filter on G_S to obtain information about the similarity between nodes (samples). By leveraging the properties of G_S and the low-pass filter, we encode the mutual correlations among samples belonging to the same fault type.

To implement the low-pass filter on the homophilic view, we utilize GNNs that can effectively capture low-frequency information in graph signals. This type of GNN acts as the low-pass filter and operates on the graph structure where similar nodes are connected. By leveraging the homophilic structure, the GNN smoothens the node features, preserving shared knowledge among similar nodes [16].

To enhance the stability and balance of the network's learning process, self-loops are introduced in the adjacency matrix before applying the low-pass filter. The adjacency matrix is then normalized using the Laplacian normalization method, expressed as $\tilde{\mathbf{A}}_S = \mathbf{D}^{-1}(\mathbf{A}_S + \mathbf{I})$, where $\mathbf{D}$ is the degree matrix of G_S graph, and $\mathbf{I}$ is the identity matrix. Subsequently, a GNN is employed to aggregate the shared features among nodes in G_S. The aforementioned process can be formalized as follows:

$$\mathbf{H}^{(s)} = \text{softmax}\left(\tilde{\mathbf{A}}_S \text{ReLU}\left(\tilde{\mathbf{A}}_S \mathbf{X}\mathbf{W}^{(0)}\right)\mathbf{W}^{(1)}\right) \quad (2)$$

where $\mathbf{H}^{(s)} \in \mathbb{R}^{n\times f}$ represents the node representation matrix of the G_S graph learned through the low-pass filter, n indicates the number of nodes in the G_S graph, and f represents the output layer's dimension. $\mathbf{W}^{(0)} \in \mathbb{R}^{d\times h}$ is the weight matrix in the low-pass filter that maps the original features from the input layer to the hidden layer feature space, $\mathbf{W}^{(1)} \in \mathbb{R}^{h\times f}$ is the weight matrix from the hidden layer to the output layer. With Rectified Linear Unit (ReLU) as the activation function, we use the softmax function for normalization.

2) High-Pass Filtering: To effectively leverage the differing information between various faults, a high-pass filter is employed on G_{DS} with edges connecting nodes that exhibit differences. This high-pass filter acts as the inverse operation of a low-pass filter and is designed to sharpen node features along the edges, thereby preserving the differing information present in the data. The high-pass filter can be formalized as follows:

$$\mathbf{H}^{(\text{ds})} = \text{softmax}\left(\left(\mathbf{I} - \tilde{\mathbf{A}}_S\right)\text{ReLU}\left(\left(\mathbf{I} - \tilde{\mathbf{A}}_S\right)\mathbf{X}\mathbf{W}^{(2)}\right)\mathbf{W}^{(3)}\right) \quad (3)$$

where $\mathbf{H}^{(ds)} \in \mathbb{R}^{n\times f}$ represents the node representation matrix learned through the high-pass filter for the G_{DS} graph, n represents the number of nodes in the G_{DS} graph. $\mathbf{W}^{(2)}$ and $\mathbf{W}^{(3)}$ are learnable parameters like $\mathbf{W}^{(0)}$ and $\mathbf{W}^{(1)}$.

Based on the findings of the Section II, we average the embeddings of $\mathbf{H}^{(s)}$ and $\mathbf{H}^{(ds)}$ to obtain the final embedding $\mathbf{H}$, where the i-th row $\mathbf{h}_i \in \mathbb{R}^f$ corresponds to the representation vector of node v_i.

B. Ranking Contrastive Learning

Based on the comprehensive analysis of the relationships between different transformer fault types discussed in Section II, it becomes evident that the boundaries between fault categories in DGA data are often ambiguous. Inspired by rank-based contrastive learning [12], we incorporate the concept of rank-based contrastive learning to penalize fault categories with a high degree of gas overlap. By incorporating this refined learning

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technique, our model can effectively distinguish between different fault types and enhance the overall performance of fault detection. The following describes this implementation in detail.

To begin with, to quantify the degree of overlap in gas characteristics between different fault categories, we calculate the similarity between fault categories, where a higher overlap corresponds to a higher similarity score. Subsequently, for each specific fault category, we rank the other fault categories in descending order of their similarity scores, thereby constructing an ordered sequence of fault categories. This process generates a ranked positive sample set denoted as $\mathcal{P}_1, \mathcal{P}_2, \ldots, \mathcal{P}_c$. Here, the variable c represents the number of fault categories resulting from the same factor. As a result of this process, we derive a set of fault categories arranged in descending order according to their degree of gas overlap.

For example, in the case of DGA-200, there are four fault categories linked to the factor of low-temperature overheating, including low-temperature overheating itself. Ultimately, each positive sample set consists of multiple samples originating from the same fault category and adheres to the following conditions:

$$s\left(\mathbf{h}, \mathbf{h}_1\right) > s\left(\mathbf{h}, \mathbf{h}_2\right) > \cdots > s\left(\mathbf{h}, \mathbf{h}_c\right) > s\left(\mathbf{h}, \mathbf{h}_n\right) \quad \forall \mathbf{h}_i \in \mathcal{P}_i,\ \mathbf{h}_n \in \mathcal{N} \tag{4}$$

where $s(\cdot)$ represents the cosine similarity used to calculate the similarity between nodes, $\mathbf{h}$ is the representation of the target node, $\mathbf{h}_i$ is the representation of a node in the positive sample set, $\mathbf{h}_n$ is the representation of a node in the negative sample set, $\mathcal{P}_i$ represents the i-th set of positive samples, and $\mathcal{N}$ represents the set of negative samples.

Afterward, we recursively employ the Infonormalized cross entropy (InfoNCE) [17] within the ranked set of fault categories, using a temperature hyperparameter to control the value of InfoNCE and increase the distance between fault categories with a high degree of gas overlap. In the initial stage, we select the first positive sample set and treat other positive sample sets as negative samples. At this point, the temperature hyperparameter is at its lowest value, and the loss function reaches its maximum. Upon completing this step, the current positive sample set is discarded, and we move to the next set. We repeat this process iteratively until there are no more positive sample sets remaining. As the process progresses, the temperature hyperparameter gradually increases, and the value of the loss function decreases accordingly. This approach incrementally addresses the issue of unclear boundaries between fault categories caused by gas overlap. More precisely, the loss function is defined as $\mathcal{L}_{\text{rank}} = \sum_i^c l_i$, where the calculation of l_i is as follows:

$$l_i = -\log \frac{\sum_{\mathbf{h}_p \in \mathcal{P}_i} \exp \frac{s(\mathbf{h}, \mathbf{h}_p)}{\tau_i}}{\sum_{\mathbf{h}_p \in \cup_{j \geq i} \mathcal{P}_i} \exp \frac{s(\mathbf{h}, \mathbf{h}_p)}{\tau_i} + \sum_{\mathbf{h}_n \in \mathcal{N}} \exp \frac{s(\mathbf{h}, \mathbf{h}_n)}{\tau_i}} \tag{5}$$

where τ_i represents temperature hyperparameter, and when $\tau_i < \tau_{i+1}$, it penalizes high similarity between faults, thereby enhancing the differentiation between fault categories with overlapping gas features.

By performing ranking contrastive learning, we can effectively eliminate noises that may exist between fault categories caused by similar factors. This approach enables the extraction of differential information between different fault categories, thereby enhancing the ability to distinguish them. Moreover, it helps to expand the fuzzy boundaries that often exist among similar faults.

C. Fault Detection

Once this refined representation is obtained, it is then utilized for transformer fault detection. This process involves refining the representation to a multilayer perceptron, which serves as a classification model. By leveraging this approach, we can effectively detect and classify transformer faults based on the learned feature representation as follows:

$$\hat{\mathbf{y}}_i = \text{softmax}\left(\mathbf{W}^{(4)}\mathbf{h}_i + \mathbf{b}\right) \tag{6}$$

where $\hat{\mathbf{y}}_i \in \mathbb{R}^C$, C represents the number of fault categories. $\hat{\mathbf{y}}_i$ signifies the probability that node i belongs to each fault category. $\mathbf{W}^{(4)} \in \mathbb{R}^{C \times f}$ is the transformation matrix for node representations. We also take the node classification cross-entropy loss as follows:

$$\mathcal{L}_{\text{main}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c} \log \hat{y}_{i,c} \tag{7}$$

where $y_{i,c}$ represents whether the true label of the ith sample belongs to category c. If it does, $y_{i,c}$ is 1; otherwise, it is 0. N represents the total number of training samples.

In the end, the learning objective balances the importance of two optimization goals, formalized as

$$\mathcal{L} = \mathcal{L}_{\text{main}} + \lambda \mathcal{L}_{\text{rank}} \tag{8}$$

where $\mathcal{L}$ is final learning objective. λ is a hyperparameter set to 0.2 in this article.

To deal with the inherent data imbalance issues, we adopt a commonly adopted loss function engineering strategy [18] for addressing class imbalance problems. The core of this strategy lies in the meticulous adjustment of the loss function, aiming to guide the model during training to allocate greater attention to minority class samples, thereby improving the overall performance of the model when dealing with imbalanced datasets. Inspired by the work in [19], we adopt the following improved loss function:

$$\mathcal{L}_{\text{imloss}} = -\frac{1}{N}\sum_{i=1}^{N} \frac{N_{\bar{c}}}{N_c} \sum_{c=1}^{C} y_{i,c} \log \hat{y}_{i,c} \tag{9}$$

where $N_{\bar{c}}$ represents the mean number of samples across all classes, N_c represents the total number of samples in class c.

IV. Experiments

In this section, we have conducted extensive experiments to evaluate the effectiveness of the proposed DGRCL. The experiments aim to address the following five research questions.

RQ1: Does DGRCL consistently outperform other baselines by yielding better performance?

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TABLE II
STATISTICS INFORMATION OF DGA-200 AND DGA-2000

Dataset	Classes	Features	Nodes	K	Edges (G_S)	Edges (G_{DS})
DGA-200	7	5	201	4	774	546
DGA-2000	7	5	1982	4	7893	5231

RQ2: Are the main components of each module work well?
RQ3: How interpretable is DGRCL?
RQ4: What is the influence of core hyper-parameters on DGRCL?
RQ5: How robust is DGRCL?

A. Experimental Setup

1) Datasets: We conduct experiments on two benchmark datasets, presenting detailed statistical information in Table II. In the table, G_S and G_{DS} graphs exclude self-loops when computing edges. Moreover, in cases where nodes i and j are mutual neighbors, the edge is counted only once.

DGA-200: The DGA-200 dataset consists of 201 samples and is composed of seven fault categories: low-temperature overheating (T1), low-to-medium temperature overheating (LMO), medium-temperature overheating (T2), high-temperature overheating (T3), PD, spark discharge (D1), and arc discharge (D2).

DGA-2000: After removing duplicates, DGA-2000 comprises 1982 valid samples. It is worth noting that spark discharge is synonymous with low-energy discharge, while arc discharge and high-energy discharge belong to the same category. Unlike DGA-200, it lacks LMO faults but includes a normal category (N).

2) Baselines and Evaluation Protocols: To assess the performance of the proposed DGRCL, we compare it with a series of benchmark methods, including traditional machine learning methods, such as SVM, KNN, BPNN, and Tree, as well as enhanced traditional methods like EBMT [20] and the GNN-based approach GraphSmin [21] as baseline methods. To provide a comprehensive evaluation of the performance of DGRCL in transformer fault detection, we have utilized commonly used metrics[20], [21]: AUC, F1 score, and precision. A higher value for all metrics implies a better model.

3) Implementation Details: DGRCL is implemented using PyTorch and trained using the Adam optimizer. To ensure a fair comparison across datasets, we performed fine-tuned hyperparameter optimization for both datasets. For DGA-200, the dropout rate is set to 0.01, the learning rate to 0.01, with a hidden layer dimension of 128 and 2 hidden layers. For DGA-2000, the dropout rate is set to 0.01, the learning rate to 0.1, with a hidden layer dimension of 256 and 4 hidden layers. The impact of key hyperparameters on model performance will be analyzed in subsection *E*. Specifically, for each experimental run, we randomly allocate 70% of each dataset to the training set, with the remaining 30% used as the test set to ensure the independence and validity of data validation. To ensure the fairness of the experiments, all comparison methods have utilized the same training and testing datasets as the DGRCL. Specifically, for GraphSmin, we have employed the GraphSmin$_1$ version.

B. Performance Comparison

This subsection first compares the performance of all methods and then explores the improvement effect of the proposed model under the setting of coarse-grained category data.

1) Overall Comparison: The overall performance comparison is located in the upper part of Table III, and we can make the following observations.

Observation 1—The proposed DGRCL achieves excellent performance: While DGRCL achieves a suboptimal F1 score on the DGA-2000 dataset, it demonstrates optimal performance across all other metrics on both datasets. Traditional machine learning methods, not designed for DGA data, generally perform poorly. KNN performs well on the DGA-2000 dataset due to its ability to accurately capture relationships between samples in large datasets. EBMT, tailored for DGA, outperforms traditional methods, while GraphSmin, addressing class imbalance in DGA using GNNs, emerges as the strongest competitor. However, DGRCL surpasses GraphSmin, highlighting the effectiveness of ranking-based contrastive loss in enhancing fault category distinction and reducing boundary ambiguity, thereby improving classification precision. *Observation 2—The AUC for all methods surpass the other two metrics:* This is due to the imbalance in the benchmark datasets, where negative samples significantly outnumber positive ones. AUC evaluates sample ranking based on predicted probabilities, making it more robust for imbalanced data. The large number of negative samples reduces precision, thereby affecting the F1. For instance, in the DGA-2000 dataset, there are only 118 positive samples for the low-temperature overheating fault, lowering both precision and F1. Given the data imbalance, AUC is prioritized in experiments. DGRCL achieves AUC improvements on both datasets, outperforming GraphSmin. This highlights the effectiveness of ranking-based contrastive learning, improving classification boundary clarity.

To thoroughly evaluate the effectiveness of DGRCL in alleviating the issue of ambiguous fault class boundaries, Table IV provides a detailed comparison of the prediction results between DGRCL and the optimal baseline method GraphSmin, on the DGA-200 test set. The value in the i-th row and j-th column represents the number of samples from the true class i that were classified as class j. A higher value along the main diagonal indicates a greater number of correctly predicted samples. From the table we can see that, for most fault classes, GraphSmin and DGRCL achieve comparable numbers of correct predictions. While DGRCL exhibits slightly lower performance than GraphSmin in predicting the D1 and D2 categories, it is noteworthy that GraphSmin fails to predict any samples in the T1 category, whereas DGRCL successfully classifies all samples in this category with complete accuracy. This significant difference highlights the superior performance of DGRCL in addressing feature overlap and ambiguous class boundaries, leading to a substantial improvement in fault diagnosis tasks.

2) Generalization Comparison: To assess the generalization ability of DGRCL on coarse-grained data, we merge the data labels from DGA-200 into temperature and discharge categories, denoted as DGA-200-2, and from DGA-2000 into normal, temperature, and discharge categories, denoted as DGA-2000-3, to

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TABLE III
PERFORMANCE COMPARISON

Methods	DGA-200			DGA-2000		
	AUC	F1	Precison	AUC	F1	Precison
BPNN	0.72±0.07	0.67±0.15	0.67±0.15	0.79±0.03	0.59±0.03	0.57±0.03
SVM	0.75±0.03	0.65±0.05	0.65±0.05	0.79±0.01	0.73±0.01	0.74±0.01
KNN	0.76±0.03	0.69±0.05	0.69±0.05	0.83±0.01	**0.79±0.02**	0.79±0.02
Tree	0.77±0.03	0.70±0.07	0.70±0.07	0.78±0.02	0.73±0.03	0.73±0.03
EBMT	0.88±0.02	0.69±0.05	0.71±0.05	0.86±0.02	0.74±0.04	0.75±0.04
GraphSmin	0.93±0.05	0.75±0.08	0.77±0.06	0.90±0.04	0.76±0.09	0.76±0.03
DGRCL	**0.94±0.02**	**0.81±0.06**	**0.82±0.05**	**0.91±0.01**	0.78±0.02	**0.79±0.01**
DGRCL-data	0.94±0.03	0.81±0.06	0.81±0.06	0.88±0.02	0.84±0.02	0.84±0.02
DGRCL-cat	0.90±0.02	0.75±0.03	0.80±0.03	0.87±0.02	0.76±0.02	0.77±0.01
DGRCL-normal	0.88±0.03	0.74±0.03	0.77±0.03	0.82±0.02	0.70±0.01	0.72±0.02
DGRCL-imloss	0.94±0.02	0.82±0.04	0.85±0.04	0.89±0.02	0.77±0.02	0.79±0.02

The optimal and suboptimal methods are highlighted in bold and underlined, respectively.

TABLE IV
DISPLAY OF PREDICTED RESULTS FOR DGRCL AND GRAPHSMIN

Methods	DGRCL							GraphSmin						
	T1	LMO	T2	T3	PD	D1	D2	T1	LMO	T2	T3	PD	D1	D2
T1	3	0	0	0	0	0	0	0	0	1	0	0	0	2
LMO	0	4	0	1	1	0	0	0	4	1	0	1	0	0
T2	0	0	4	1	0	0	0	0	1	4	0	0	0	0
T3	0	0	0	15	0	0	0	0	0	0	15	0	0	0
PD	0	1	1	0	3	0	0	1	1	0	0	2	0	1
D1	0	0	0	3	0	13	1	0	0	0	0	0	16	1
D2	0	0	0	0	1	1	10	0	0	0	0	0	0	12

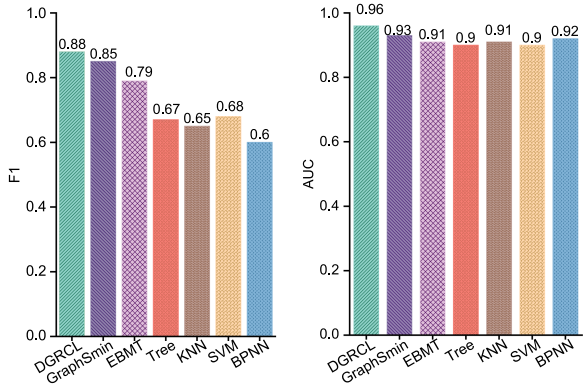


Fig. 3. Generalization of DGRCL. The left figure compares the F1 performance of DGRCL and baselines on DGA-200-2, while the right compares their AUC on DGA-2000-3.

construct coarse-grained data. The left side of Fig. 3 presents the F1 scores of each model on the DGA-200-2 dataset, while the right side displays the AUC values on the DGA-2000-3 dataset. We make the following observations.

Observation 1—DGRCL can effectively expand fault boundaries: On the DGA-200-2 dataset, which lacks a normal category and has relatively ambiguous class boundaries, DGRCL outperforms the best benchmark(GraphSmin) in terms of the F1 score by 3% . This indicates that DGRCL effectively expands the fuzzy boundaries between fault categories, thereby enhancing the accuracy of transformer fault diagnosis. *Observation 2—DGRCL is capable of capturing pivotal features:* In the DGA-2000-3 dataset, where a normal category is present and class boundaries are relatively distinct, all methods achieve AUC values exceeding 90% . However, DGRCL still outperforms the GraphSmin, demonstrating its ability to capture key features from the samples to maintain model performance, particularly in scenarios with pronounced class differences. This highlights DGRCL's superior generalization capability on coarse-grained data.

3) Efficiency Analysis: To comprehensively demonstrate the effectiveness of DGRCL, we not only compare its performance with baseline methods but also further validate its efficiency from the perspective of spatiotemporal complexity. Theoretically, DGRCL constructs the KNN graph with a time complexity of $O(n^2\ (d+\log n))$, where n is the number of samples and d is the feature dimension. The graph filtering has an approximate time complexity of $O(md+mh)$, where m is the number of nonzero elements in adjacency matrix and h represents the hidden layer feature dimension. The ranking-based contrastive loss component involves similarity calculations with a time complexity of $O(n^2 f)$, where f denotes the final embedding dimension. Although the construction of the KNN graph entails a high computational complexity, this is a one-time preprocessing step

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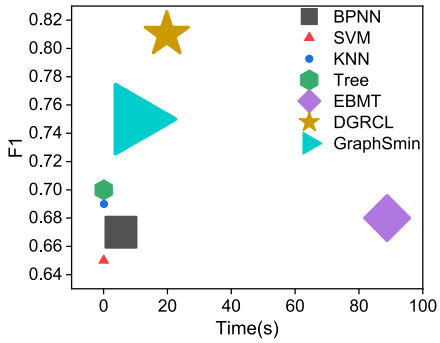


Fig. 4. Spatiotemporal efficiency on DGA-200.

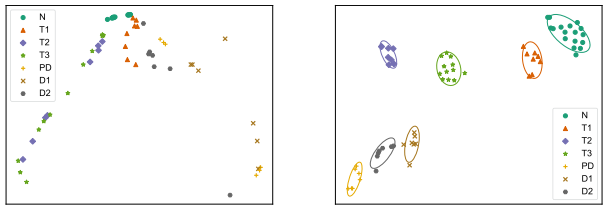


Fig. 5. Visualization on original embedding and learned embedding space on the DGA-2000 test set. (a) Original feature. (b) Learned embeddings.

and is not part of the model's training process. Consequently, the overall time complexity of DGRCL is $O(md + mh + n^2 f)$.

Fig. 4 offers a comparative evaluation of the DGRCL against six baseline models on the DGA-200 dataset, with an emphasis on training runtime, parameter quantity, and fault diagnosis accuracy. The x-axis indicates the training time, the y-axis measures the F1 score, and the size of the markers corresponds to the relative number of parameters. Models positioned in the upper left quadrant with smaller marker sizes denote superior overall efficiency and accuracy.

As depicted in Fig. 4, the majority of traditional machine learning-based approaches have a lower parameter count but deliver subpar performance. Despite having a similar number of parameters as DGRCL, both BPNN and the enhanced machine learning method EBMT, fail to demonstrate substantial performance gains, suggesting that an increase in parameter count alone does not contribute to better model performance. Besides, GraphSmin, which employs dual-channel graph representation learning and standard contrastive learning techniques, still underperforms compared to DGRCL. Overall, DGRCL achieves the highest fault diagnosis accuracy while maintaining a manageable level of computational complexity. This underscores the efficacy of ranking-based contrastive learning in diminishing the boundary ambiguity for fault classification.

C. Ablation Study

To address RQ2, we conduct ablation experiments from three perspectives: data-level (DGRCL-data), model-level (DGRCL-cat, DGRCL-normal) and loss-level (DGRCL-imloss). The data-level ablation experiments aim to evaluate the performance of DGRCL in practical application scenarios, where specific class data may be missing due to operational conditions, such as equipment overload or data collection biases. The model-level ablation experiments are conducted to verify the effectiveness of each component of DGRCL, ensuring that every module functions as intended. Lastly, the loss-level ablation experiments are designed to provide insights for future research, exploring more effective strategies for adjusting and optimizing loss functions when addressing data imbalance issues. Specifically, we perform three ablation experiments: *DGRCL-data* retains only temperature-induced fault data from DGA-200 and discharge-induced fault data from DGA-2000. *DGRCL-cat* simplifies the representation of the ranking contrastive learning module by using a concatenation approach. *DGRCL-normal* replaces ranking contrastive learning in DGRCL with regular contrastive learning [22]. *DGRCL-imloss* uses the loss formula (9) to replace the formula (7). Ablation experiments related to the dual-channel graph representation module have been extensively discussed in prior work and have been adopted based on prereserach results to avoid redundancy. The lower part of Table III presents the results of various ablations of DGRCL. We make the following observations: *Observation 1—DGRCL demonstrates robustness:* Even in scenarios with partial data loss, DGRCL maintains stable performance, indicating its high resilience to data incompleteness. This suggests that the model can effectively preserve its reliability in practical applications. *Observation 2—Fuzzy features impact performance:* In the model-level ablation experiments, DGRCL performs well on DGA-200 but poorly on DGA-2000 due to lower fault similarity in DGA-200. In ranking contrastive learning, lower similarity aids boundary expansion, while higher similarity makes it challenging. *Observation 3—Ranking contrastive loss is effective:* DGRCL outperforms DGRCL-normal and DGRCL-cat, indicating the effectiveness of ranking contrastive loss in delineating fuzzy boundaries. DGRCL-cat outperforms DGRCL-normal, suggesting simple contrastive learning confuses the model with high similarity between different faults. *Observation 4—Loss optimization mitigates data imbalance:* DGRCL-imloss achieves performance improvements on the DGA-200 dataset, although its performance declines on the DGA-2000 dataset. These results indicate that loss engineering can effectively alleviate mild data imbalance issues. Furthermore, this implies that there is significant potential for further optimization and exploration of solutions to data imbalance problems in future research.

D. Visualization

To investigate the effectiveness of DGRCL in addressing the issue of fuzzy boundaries, we randomly selected samples from the DGA-2000 test set and utilized t-distributed Stochastic Neighbor Embedding (t-SNE) for visualization. Fig. 5(a) shows the t-SNE visualization of the original features, while Fig. 5(b) presents the feature visualization after DGRCL learning. Based on the observations from Fig. 5, we draw the following conclusions:

Observation 1—Fault clustering: DGRCL enables the formation of distinct clusters for each category, which strongly

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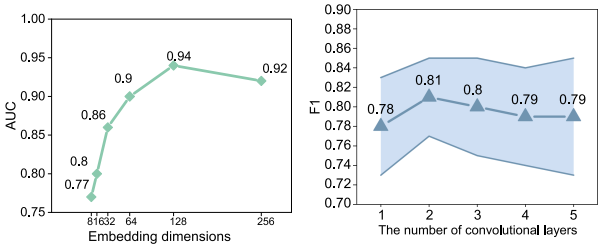


Fig. 6. Analysis of hyperparameters on the DGA-200 dataset. (a) AUC versus embedding dimension. (b) F1-score versus number of convolutional layers.

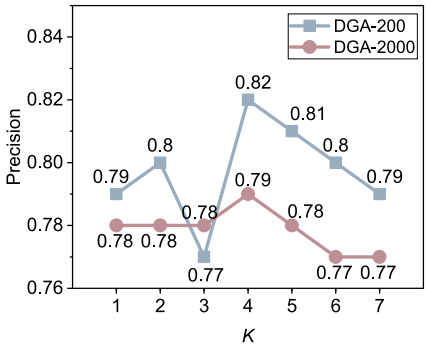


Fig. 7. Performance variation under different K values.

indicates that DGRCL effectively learns shared attributes within fault categories, thereby enhancing the clustering effect in the feature space. *Observation 2—Boundary expansion:* DGRCL successfully distinguishes the nearly overlapping T2 and T3 categories in the original space, demonstrating its ability to uncover and learn the latent differences between fault categories, further clarifying the interclass boundaries.

E. Hyperparameter Analysis

In this section, we first study the impact of the hidden layer embedding dimension and the number of convolutional layers on the model using the DGA-200 dataset, as shown in Fig. 6. Fig. 6(a) illustrates the variation of the AUC with the hidden layer embedding dimension, while Fig. 6(b) displays the changes in the F1 score with the number of convolutional layers. The colored bands in Fig. 6(b) represent the standard deviation range of the results from 10 random splits of the dataset. As both hyperparameters increase, the performance of DGRCL initially increases and then decreases. Specifically, the model achieves optimal performance when the embedding dimension is 128, while for the number of convolutional layers, the best performance is observed when the number of layers is 2. This is because excessively large hyperparameters tend to lead to overfitting, resulting in a gradual decline in performance and an increase in the error range.

We further investigate how the key hyperparameter K affects the performance of DGRCL, Fig. 7 depicts the impact of varying K values on model performance for both the DGA-200 and DGA-2000 datasets, with all other hyperparameters remaining constant. Fig. 7 reveals that the performance of both datasets

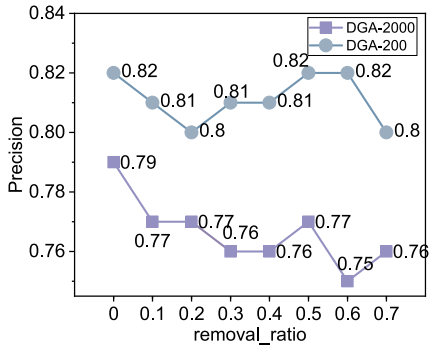


Fig. 8. Performance variations of DGRCL under simultaneous random edge removal in both similar and dissimilar graphs.

reaches its optimum at $K = 4$, suggesting that DGRCL effectively captures the local structural information of the samples at 4. Additionally, the DGA-2000 dataset exhibits low sensitivity to variations in K, likely due to its relatively large sample size. This large sample size leads to more complex and densely interconnected relationships among the samples, thereby increasing the model's robustness to changes in K.

F. Robustness Analysis

To evaluate the robustness of the DGRCL, this section introduces random perturbations to the edges in both similarity and dissimilarity graphs to simulate scenarios where data structures are damaged or incomplete, reflecting real-world challenges. The performance of DGRCL under these conditions is observed to examine its stability when exposed to varying degrees of structural disruption. Specifically, edges in the similarity and dissimilarity graphs are randomly removed at proportions of $\{0.1, 0.2, 0.3, 0.4, 0.5, 0.7\}$ to simulate different levels of structural damage. The experimental results are presented in Fig. 8.

As observed in Fig. 8, the performance of DGRCL exhibits a decline under varying levels of edge removal perturbations; however, the degree of decline remains relatively minor, demonstrating the model's robustness against disturbances. Under conditions of relatively high perturbation (e.g., 30% –50%), the performance of DGRCL shows minimal fluctuation and may even exhibit slight improvement, attributed to its fault-tolerant capabilities. Even in extreme scenarios, such as a 70% edge removal ratio, the model's performance across two datasets remains within an acceptable range. This indicates that DGRCL is capable of effectively handling incomplete or noisy data in real-world applications. Overall, the stable performance of the DGRCL model under various perturbation levels highlights its robustness in dealing with complex disturbances.

V. Conclusion and Future Work

This article presents a graph contrastive learning model, DGRCL, based on DGA data for power transformer fault detection. DGRCL utilizes a dual-channel GNN to capture contextual information among transformer samples and employs ranking contrastive learning to comprehensively understand fault patterns in transformer data. DGRCL holds significant application potential in practical settings. For instance, it can

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be integrated into monitoring and maintenance platforms within power systems to continuously track the operational status of transformers, thereby enabling the early detection of potential faults and reducing the risk of unexpected power outages and equipment damage. In real-world applications, challenges the model may face include efficiently processing large volumes of real-time data to ensure timely and effective detection, as well as adjusting model hyperparameters to accommodate different types of transformers and fault patterns for various operating scenarios.

It is worth noting that the imbalance in DGA data restricts the performance of power transformer fault detection. Therefore, in future work, a combined approach using GNNs can be explored to address the issue of imbalance in DGA data for transformer fault detection based on dissolved gases in oil.

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